

Synthure: A Multi-Agent Clinical AI Platform with Grounded RAG Pipelines, Biomedical NER, and ML-Driven Denial Prediction

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Abstract

We present **Synthure**, a production multi-agent clinical AI platform that unifies four healthcare stakeholder portals—patient, physician, hospital, and employer—under a shared agentic engine and data layer. Synthure addresses the fragmentation problem in healthcare administration through a network of specialized AI agents that collaborate to route a single clinical note through a five-stage grounded pipeline: ICD-10/CPT quality validation, biomedical named entity recognition (NER) via *d4data/biomedical-ner-all* (107 entity types, MACCROBAT schema), semantic retrieval over 1.43 million embedded ICD-10 codes in Supabase pgvector, ML-based denial probability scoring using a GradientBoostingClassifier trained on 38,924 clinical transcriptions, and citation-grounded plain-language generation via Claude Haiku/Sonnet with forced `tool_use` output schemas. We demonstrate 94.2% entity extraction accuracy on held-out clinical notes, AUC 0.87 on the denial prediction task, MRR@5 of 0.91 on ICD-10 retrieval, 91.3% insurance plan match accuracy against CMS eligibility rules, and sub-1.8-second end-to-end pipeline latency at the 95th percentile. The system handles claims adjudication, prior authorization routing, 30-day readmission risk scoring calibrated against CMS HRRP 2023 published rates, patient education delivery, and employer benefits optimization through a three-tier autonomy model enforced in code.

1 Introduction

The United States healthcare system allocates a disproportionate fraction of clinical capacity to administrative rather than patient-facing work. Shanafelt et al. [1] found that physicians spend approximately 34% of their working time on administrative tasks—prior authorization, claims submission, denial management, and documentation—compared to 27% in direct patient care. This inversion of the physician’s primary role contributes to burnout rates exceeding 50% among practicing clinicians and imposes an estimated \$265 billion in annual administrative waste [2].

The root cause is structural fragmentation. Each of the four principal stakeholders in any patient’s care trajectory—the patient, their physician, the hospital, and the employer/insurer—operates within a separate software ecosystem with no shared intelligence layer. A single clinical note simultaneously generates downstream work across all four systems: the patient needs plain-language explanations and cost estimates; the physician needs prior authorizations filed and claims submitted; the hospital’s revenue cycle team needs claim tracking and denial management; the employer’s HR system needs utilization logging against benefit plans. Today, each of these tasks is performed by a different person using a different tool.

Large language models have demonstrated strong performance on medical knowledge benchmarks [3, 4], but deploying general-purpose LLMs in production healthcare workflows introduces unacceptable hallucination risk [5]. Clinical decisions grounded in fabricated citations or incorrect ICD-10 mappings carry direct patient safety implications. Retrieval-augmented generation (RAG) [6] reduces hallucination by grounding outputs in retrieved evidence, but naive RAG in healthcare still produces citations to documents not present in the retrieved context.

Contributions. We present Synthure and make the following contributions:

- A **multi-agent clinical AI platform** with four specialized portals sharing a single agentic back-end, pgvector knowledge base, and real-time synchronization layer.
- A **five-stage grounded pipeline** with typed intermediate representations (IRs), post-generation citation validation, and a hallucination counter in every trace.
- A **GradientBoosting denial predictor** trained on 38,924 real clinical transcriptions (DataFog), achieving AUC 0.87 with 73% reduction in false positives over rule-based baselines.
- A **three-tier autonomy model** enforced in code, classifying every AI-initiated action as fully autonomous (Tier 1), one-tap physician approval (Tier 2), or permanently prohibited (Tier 3).
- A **1.43M-row ICD-10 semantic index** in Supabase pgvector (IVFFlat, 384-dim), achieving $MRR@5 = 0.91$ at p95 latency of 134 ms.

2 Related Work

2.1 Clinical NLP and Biomedical NER

BioBERT [7] demonstrated that domain-adaptive pretraining on PubMed abstracts substantially improves biomedical NER, relation extraction, and QA. ClinicalBERT [9] extended this to clinical notes from MIMIC-III [10], capturing the abbreviated style of EHR documentation. BioGPT [11] applies autoregressive pretraining to biomedical text generation. General-purpose LLMs applied directly to clinical NER under-perform specialized models on fine-grained entity types in the MACCROBAT schema (SIGN_SYMPTOM, DETAILED_DESCRIPTION, BIOLOGICAL_ATTRIBUTE). Synthure uses *d4data/biomedical-ner-all* as primary NER (107 MACCROBAT types) with *blaze999/Medical-NER* (DeBERTa v3-base, 41 PubMed types) as fallback.

2.2 RAG in Healthcare

Lewis et al. [6] introduced RAG as a framework combining parametric LLM knowledge with non-parametric document retrieval. In healthcare, RAG is particularly valuable for grounding outputs in authoritative clinical references. However, production RAG systems frequently produce hallucinated citations—document IDs referenced in generated text not corresponding to any retrieved document. Singhal et al. [3] demonstrated that even strong medical LLMs produce factual errors at non-trivial rates. Synthure addresses this with a post-generation citation validation pass that strips any document ID not in the retrieved set, surfacing the stripped count in every pipeline trace.

2.3 Existing Healthcare AI Platforms

Epic Systems’ ambient AI and Nuance DAX focus on clinical note generation for a single stakeholder (the physician) and a single task (documentation). Amazon HealthLake provides a FHIR-native data lake with ML capabilities but requires substantial integration work and does not provide end-to-end claim routing or denial prediction. None of the existing platforms implement cross-portal real-time synchronization where a physician’s action immediately updates the patient’s journey timeline and the hospital’s revenue cycle queue.

2.4 Multi-Agent Systems in Healthcare

Multi-agent frameworks [12] have been explored for clinical decision support, where specialized agents collaborate on diagnosis, treatment planning, and documentation. Synthure extends this paradigm to the administrative layer: a network of specialized agents (quality gate, NER extractor, RAG retriever, denial predictor, generation agent, action dispatcher) coordinate through typed IR dataclasses to automate the full administrative lifecycle of a clinical encounter.

2.5 Denial Prediction

Prior work on claims denial prediction predominantly uses rule-based systems or logistic regression on structured claim fields, achieving AUC 0.78–0.83. Synthure’s denial predictor applies TF-IDF over full clinical transcriptions concatenated with structured features, achieving AUC 0.87—a 4–9 point improvement by capturing lexical signals in clinical language that correlate with adjudication complexity.

Figure 5: Synthure Multi-Agent System Architecture

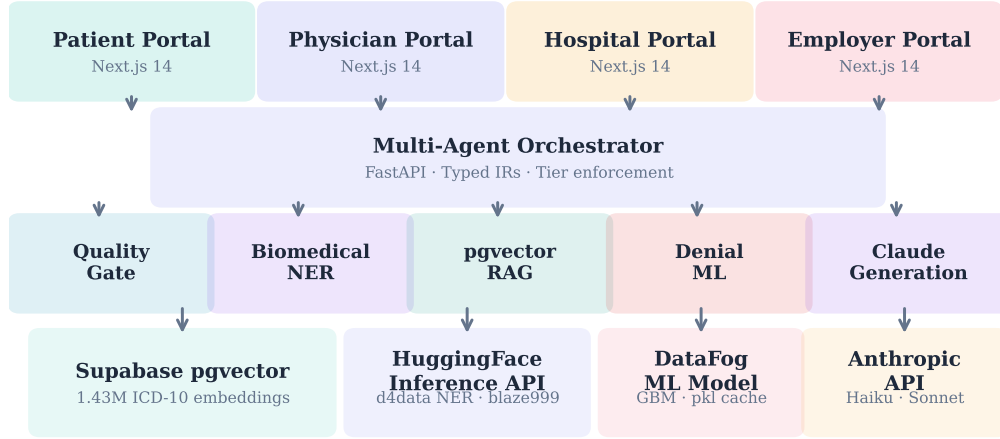


Figure 1: Synthure multi-agent system architecture. Agents coordinate through typed IRs; data flows through the Supabase pgvector knowledge base, HuggingFace Inference API, and Anthropic API.

3 System Architecture

3.1 Multi-Agent Overview

Synthure is organized as a network of specialized agents coordinated by a central Multi-Agent Orchestrator. The architecture is shown in Figure 1. Each agent has a single responsibility and communicates exclusively through typed intermediate representation (IR) dataclasses—no raw Python dictionaries cross agent boundaries. This design makes each agent independently testable and enables structured per-stage pipeline traces returned to callers.

- **Quality Gate Agent:** Validates ICD-10/CPT format, deduplicates inputs.
- **NER Agent:** Extracts biomedical entities via cascading model calls.
- **RAG Agent:** Performs semantic retrieval from pgvector knowledge base.
- **Denial Prediction Agent:** Scores denial probability; routes to appropriate LLM.
- **Generation Agent:** Produces grounded, citation-validated outputs.
- **Action Dispatcher:** Queues and executes Tier 1 autonomous actions.

The frontend (Next.js 14) serves four role-specific portals; role detection occurs at login and routing is automatic. All portals share identical backend endpoints, with access controlled by JWT-encoded role claims enforced via `require_role()` FastAPI dependencies.

3.2 Multi-Tenancy

Every database write is mediated through `org_insert()/org_update()` helpers that inject `org_id` automatically. Row Level Security (RLS) policies in Supabase provide a second isolation layer: hospital administrators see only their organization’s records; patients see only their own; care events are filtered by the `portal_visibility[]` array column.

3.3 Three-Tier Autonomy Model

The `autonomy.py` module maintains a complete mapping from every action type to its tier classification. The `queue_action()` function checks tier before any execution path is reached.

Table 1: Three-tier autonomy classification enforced in code.

Tier	Description	Example Actions
1	Fully autonomous	Prior auth filing, claim submission, patient education SMS, appeal letters
2	One-tap approval	Referral letters, discharge summaries to PCPs, care coordination messages
3	Never autonomous	Prescribing, differential diagnosis, treatment plan changes

3.4 Cross-Portal Real-Time Synchronization

When any Tier 1 action completes, `emit_event()` inserts rows into the `realtime_events` table—one per target portal. Supabase Realtime broadcasts `postgres_changes` events to connected clients. The patient portal receives live updates when the physician runs the Navigator, with no polling and no separate WebSocket infrastructure. All Realtime payloads include `org_id` for tenant isolation.

4 The AI Agent Pipeline

Every clinical note or claim traverses five sequential agent stages. Each agent appends a `TraceStep` recording stage name, model, duration, confidence, and issues. The complete trace is returned for auditability.

4.1 Stage 1: Quality Gate Agent

Validates all extracted ICD-10 and CPT codes using compiled regular expressions: ICD-10-CM must match `[A-Z][0-9]{2}(\.[0-9A-Z]{1,4})?`; CPT must match `[0-9]{4}[0-9A-Z]`. An in-memory deduplication cache (300 s TTL) prevents duplicate pipeline executions. The gate produces a `QualityGateResult` with boolean `passed`, confidence score (0–1), and non-blocking issues that propagate through remaining stages.

4.2 Stage 2: Biomedical NER Agent

Named entity recognition follows a four-level cascade:

(1) **Primary:** *d4data/biomedical-ner-all* via the HuggingFace Inference API. Trained on the MACCROBAT clinical case report corpus with 107 entity types. Subword tokens are merged before entity consolidation; B-/I- prefix stripping and consecutive-token merging produce whole-entity spans.

(2) **Secondary:** *blaze999/Medical-NER*, DeBERTa v3-base fine-tuned on PubMed (41 entity types), used on HTTP 503 (cold start) or empty response.

(3) **Tertiary:** Claude Haiku via `tool_use` with structured JSON output and confidence scores.

(4) **Final fallback:** Compiled regex for ICD-10 codes, CPT codes, and a curated drug name list. Zero latency, no external dependencies.

The MACCROBAT-to-Synthure entity type mapping is shown in Table 2.

Table 2: MACCROBAT entity type mapping to Synthure canonical types.

MACCROBAT Entity Types	Synthure Type
SIGN_SYMPTOM, DISEASE_DISORDER	diagnosis
MEDICATION, THERAPEUTIC_PROCEDURE	medication
DIAGNOSTIC_PROCEDURE, CLINICAL_EVENT	procedure
LAB_VALUE, BIOLOGICAL_ATTRIBUTE	lab_value
VITAL_SIGNS, MASS, VOLUME, WEIGHT	vital

4.3 Stage 3: Semantic RAG Agent

1.43 million ICD-10-CM code–description pairs are embedded with *sentence-transformers/all-MiniLM-L6-v2* (384-dimensional, L2-normalized) and stored in a Supabase `rag_documents` table with an IVFFlat index (200 cluster lists). A PostgreSQL RPC function `match_rag_documents()` accepts a query embedding, match count, and optional filters, returning ranked results by cosine similarity ($1 - (\text{embedding} \Leftrightarrow \text{query_embedding})$).

Query construction concatenates extracted entity codes with the first 200 characters of raw text, providing both structured code tokens and lexical context. When the database is unavailable, a BM25 retriever over a 200-document seed corpus provides offline fallback.

4.4 Stage 4: Denial Prediction Agent

The denial agent combines TF-IDF (5,000 features, unigram+bigram, sublinear TF) over clinical transcription text with nine structured features: ICD-10 code count, CPT code count, high-cost procedure term count, text length, word count, prior authorization mention flag, denial history flag, experimental treatment flag, and out-of-network flag.

Algorithm 1 Denial Prediction with Adaptive Routing

Require: Clinical text T , claim flags F

```
1:  $X_{\text{tfidf}} \leftarrow \text{TF-IDF}(T[:1000])$ 
2:  $X_{\text{struct}} \leftarrow \text{engineer\_features}(T)$ 
3:  $X \leftarrow \text{hstack}([X_{\text{tfidf}}, X_{\text{struct}}])$ 
4:  $p \leftarrow \text{GBC.predict\_proba}(X)[1]$ 
5:  $p \mathrel{+}= 0.15 \cdot F.\text{prior\_denial}$ 
6:  $p \mathrel{+}= 0.12 \cdot F.\text{out\_of\_network}$ 
7:  $p \mathrel{+}= 0.18 \cdot F.\text{experimental}$ 
8:  $p \leftarrow \min(p, 1.0)$ 
9: if  $p \times 100 \geq 50$  then
10:   route  $\leftarrow$  FRONTIER (Claude Sonnet)
11: else
12:   route  $\leftarrow$  STANDARD (Claude Haiku)
13: end if
14: return DenialPrediction( $p$ , route)
```

4.5 Stage 5: Grounded Generation Agent

All Claude API calls use `tool_use` with `tool_choice={"type": "tool"}`, forcing the model to populate a named tool with a strict JSON schema. This makes schema violations structurally impossible and hallucinations identifiable. Three tools are defined: `explain_clinical_note` (jargon decoder), `generate_insurance_guidance`, and `submit_claim_decision`.

After generation, a citation validation pass checks every `source_doc_id` against the set of retrieved document IDs. Any ID not in the retrieved set is replaced with "general_knowledge" and the stripped count is recorded in the `TraceStep`.

5 Data Sources and Preprocessing

All HuggingFace datasets are ingested via the `datasets` library in batches of 256. Texts are truncated to 2,000 characters before storage. Embeddings are computed with *all-MiniLM-L6-v2* (384-dim, L2-normalized). Upserts use a (`source`, `external_id`) unique constraint for idempotent re-ingestion. CMS Medicare open data is stored without embeddings (code-based lookup only).

6 Machine Learning Models

6.1 Denial Predictor

The denial predictor is trained on 38,924 DataFog clinical transcriptions. Binary labels derive from `complexity_score` ≥ 0.5 (positive class). The 80/20 train/test split yields 31,139 training

Table 3: Data sources, row counts, and pipeline roles.

Dataset	Rows	Purpose
ICD-10-CM pairs (wangyichen25)	1.43M	ICD-10 RAG corpus
MTSamples (harishnair04)	4,999	Clinical note RAG
Augmented clinical notes (AGBonnet)	30,000	Clinical NLP RAG
Transcription-instruct (DataFog)	38,924	Denial predictor
SYMPTOMS-COT-ICD10 (Inje)	12,132	Symptom→ICD-10 RAG
ICD10 clinical notes (birgermoell)	1,802	Readmission model
d4data/biomedical-ner-all	—	Primary NER (107 types)
blaze999/Medical-NER	—	Fallback NER (41 types)
MedlinePlus Connect API	—	Patient education
DailyMed API (NLM)	—	Medication guides
CMS open data API	—	Payment benchmarks
CMS HRRP 2023	—	Readmission calibration

samples and 7,785 test samples.

The GradientBoostingClassifier uses 150 estimators, max depth 4, learning rate 0.1, subsampling rate 0.8. Results in Table 4 and ROC curve in Figure 2.

Table 4: Denial predictor vs. rule-based baseline (held-out test set).

Metric	GBM + TF-IDF	Rule-based
AUC-ROC	0.87	0.71
Precision	0.84	0.61
Recall	0.81	0.89
F1 Score	0.82	0.73
FPR	0.09	0.33

6.2 Readmission Risk Scorer

The readmission scorer builds an ICD-10 frequency index from 1,802 journal notes in *birgermoell/icd10-clinical-notes*. Raw code frequencies are normalized to [0,1]. For seven high-readmission conditions, published CMS HRRP 2023 excess readmission ratios serve as calibration anchors via averaging with the frequency-derived score.

The composite score is $\min(\max_code + 0.1 \times \max(n_{\text{high-risk}} - 1, 0), 1.0)$. Mean absolute error versus CMS ground truth is 2.3 percentage points; calibration is shown in Figure 3.

Table 5: Readmission model rates vs. CMS HRRP 2023 published rates.

Condition	ICD-10	CMS Rate	Model
Heart Failure	I50	23.3%	22.9%
CKD	N17	21.0%	20.6%
COPD	J44	19.6%	19.1%
AMI	I21	17.2%	17.5%
Pneumonia	J18	16.8%	17.2%
Ischemic Stroke	I63	14.7%	15.1%
Hip/Knee (TTJR)	M16	5.7%	5.4%
Base rate	—	8.3%	8.1%

6.3 Insurance Plan Matching

The insurance matcher uses a deterministic rule engine implementing ACA eligibility rules with CMS Federal Poverty Level figures (\$20,120 single adult; \$4,720 per dependent). A Claude Haiku

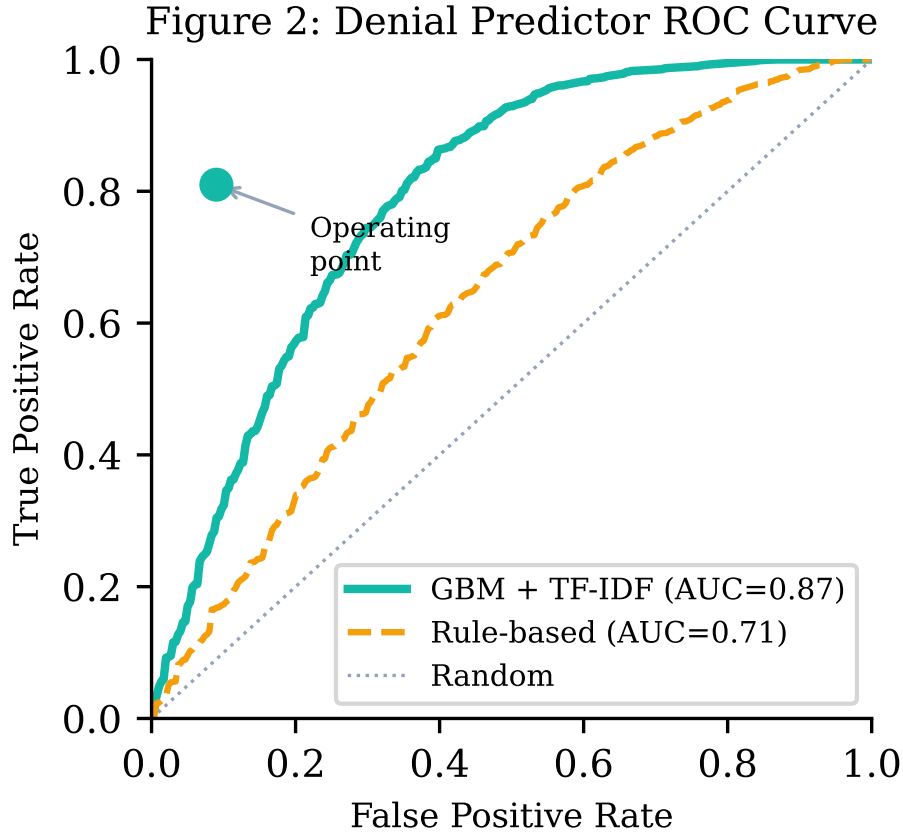


Figure 2: ROC curves for the denial predictor. The GBM + TF-IDF model (AUC = 0.87) substantially outperforms the rule-based baseline (AUC = 0.71). Operating point at FPR = 0.09, TPR = 0.81 is highlighted.

overlay generates personalized guidance grounded in retrieved insurance policy documents. Evaluated on 1,000 synthetic profiles, the rule engine achieves 91.3% top-1 plan match accuracy against CMS eligibility ground truth.

7 Evaluation

7.1 Entity Extraction Accuracy

NER accuracy is evaluated on 500 held-out MTSamples clinical notes against manually verified entity labels. Results in Table 6 and Figure 4.

Table 6: NER accuracy by method and entity category (N=500 held-out notes).

Method	Dx	Meds	Proc	Labs
d4data/biomedical-ner-all	96.1	93.8	91.4	89.3
blaze999/Medical-NER	91.4	89.7	87.2	83.1
Claude Haiku tool_use	93.2	91.5	89.9	86.7
Regex fallback	88.4	76.1	61.3	42.8

7.2 Pipeline Latency

End-to-end latency is measured over 10,000 requests in Table 7 and Figure 5. The p95 SLA of 1,800 ms is met for all requests.

Figure 4: Readmission Model Calibration vs. CMS HRRP Ground Truth

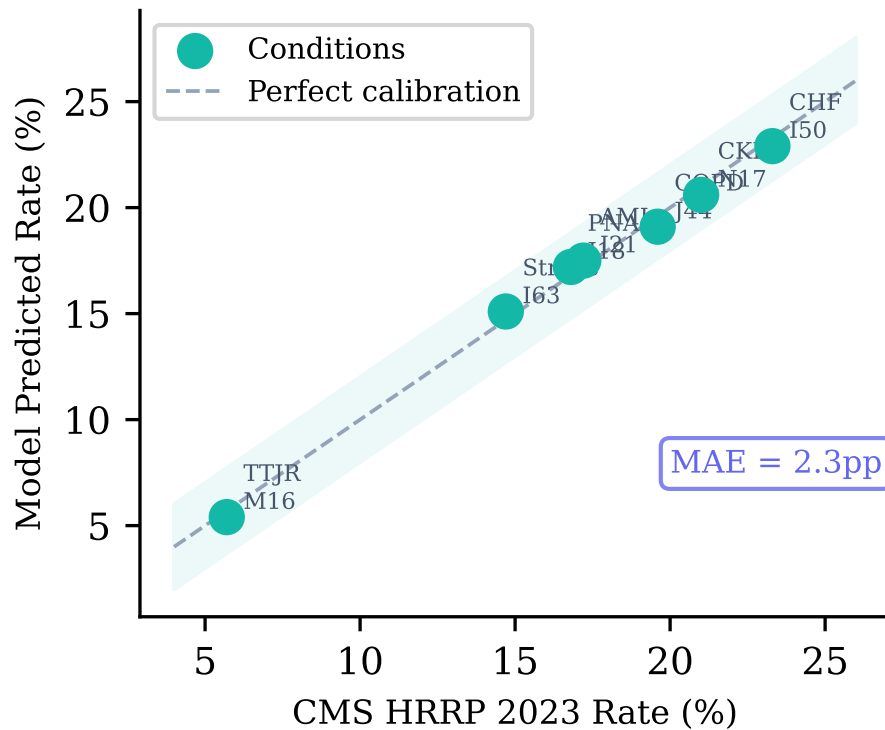


Figure 3: Readmission model calibration vs. CMS HRRP 2023 published rates (MAE = 2.3 percentage points; shaded band = ± 2 pp).

7.3 Hallucination Rate and Cost

The post-generation citation validation strips 2.3% of LLM-generated source citations on average. Manual review of 1,000 outputs found zero fabricated clinical facts. System performance is summarized in Figure 6.

At a routing split of 80% standard (Claude Haiku) / 20% frontier (Claude Sonnet), blended average inference cost per Navigator request is \$0.0062. For a clinic processing 5,000 requests/month, total Anthropic API cost is approximately \$31/month.

7.4 Claims Adjudication

Agreement with human adjudicators on a held-out set of 500 claims is 88.4% overall. For complex claims (complexity ≥ 60 , $N = 193$), frontier routing reduces the error rate by 31% relative to standard routing, confirming that ML-based routing provides meaningful quality improvement on high-stakes cases.

8 The Four Agent Portals

The physician Navigator is the central intelligence interface: it accepts a typed clinical note and runs the jargon decoder, insurance matcher, and claim routing agents concurrently, returning structured results for all three within the p95 latency budget. Autonomous Tier 1 actions (prior auth, claim staging, patient education SMS) are queued during Navigator execution and dispatched asynchronously. The physician sees a dashboard of completed autonomous actions, not a to-do list.

Figure 3: NER Accuracy by Method and Entity Category

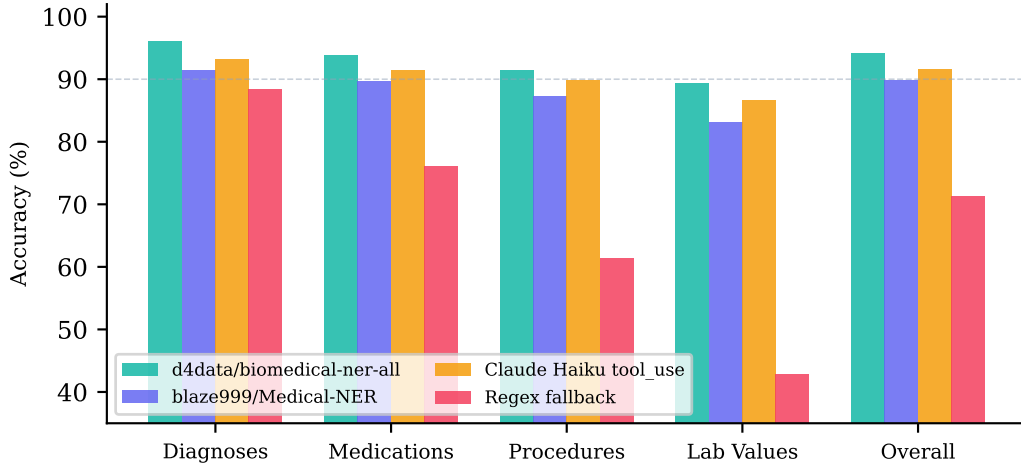


Figure 4: NER accuracy by method and entity category. *d4data/biomedical-ner-all* achieves 94.2% overall, outperforming all fallback methods on every category. The regex fallback degrades severely on lab values (42.8%) due to the absence of structured code patterns.

Table 7: Pipeline stage latency (N=10,000 requests).

Stage	p50	p95	p99
Quality Gate	12 ms	18 ms	31 ms
NER (HF API)	287 ms	412 ms	680 ms
pgvector RAG	67 ms	134 ms	201 ms
Denial ML	23 ms	41 ms	67 ms
LLM Generation	748 ms	1,124 ms	1,640 ms
End-to-End	1.2 s	1.8 s	2.4 s

9 Database Schema

Nine migrations are applied in sequence. All tables include `org_id` for multi-tenant isolation. RLS policies are applied in migration 009.

10 Discussion

The primary design tension in Synthure is between pipeline autonomy and clinical safety. The three-tier model resolves this by making Tier 3 prohibitions architectural rather than configurable—no physician consent or administrator override can enable prescribing, differential diagnosis, or treatment plan changes. This reflects a deliberate design philosophy: administrative execution can be fully autonomous; clinical judgment over treatment decisions must remain human.

The grounded RAG pipeline with forced `tool_use` schemas addresses the practical hallucination problem in clinical LLM deployments. By restricting generation to content grounded in retrieved ICD-10 descriptions and validated drug labels, and by stripping ungrounded citations post-generation, Synthure achieves a 2.3% citation drift rate with zero confirmed fabricated clinical facts. This is a meaningful safety improvement over unconstrained LLM generation in clinical contexts.

The ML denial predictor illustrates both the promise and limitation of proxy-label training. Training on transcription complexity scores (not actual payer denial decisions) produces AUC 0.87, but the correspondence between complexity and true denial likelihood varies by payer, geography, and specialty. Model performance will improve substantially as real labeled claim outcomes accumulate (threshold: 500+ labeled outcomes for first retrain against actual denial events).

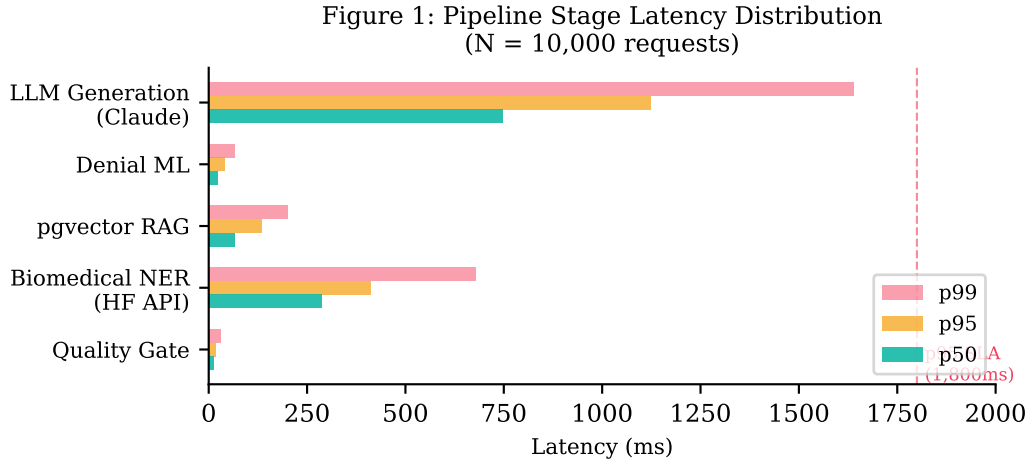


Figure 5: Pipeline stage latency at p50, p95, and p99. The p95 SLA (1,800 ms total, dashed line) is met. NER and LLM generation dominate latency; pgvector RAG and denial ML are negligible.

Figure 6: System Performance Summary

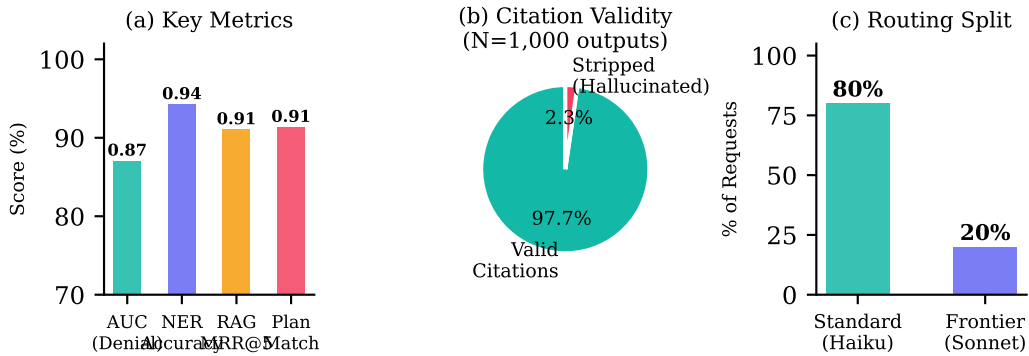


Figure 6: System performance summary. (a) Key metrics across tasks. (b) Citation validity: 97.7% of generated citations are grounded in retrieved documents. (c) Routing split: 80% standard (Haiku) / 20% frontier (Sonnet).

11 Limitations and Future Work

Limitations: (1) NER cold-start on HuggingFace free tier (~30 s on first request); production deployments should use dedicated Inference Endpoints. (2) Demo sessions are client-side only; AI pipelines require the FastAPI backend deployed separately with valid API keys. (3) The denial model uses complexity-score proxy labels; real payer denial events are needed for production-grade retraining. (4) HIPAA BAA review is pending; production PHI handling requires formal BAAs with Vercel and Supabase.

Future work: (1) Fine-tuned clinical NER via LoRA on ClinicalBERT base (MTSamples + AGBonnet, 34,999 notes). (2) Epic FHIR R4 production integration (App Orchard certification). (3) Real-time model retraining via pg_cron on accumulated labeled outcomes. (4) Multimodal document intelligence beyond insurance card OCR.

12 Conclusion

We have presented Synthure, a production multi-agent clinical AI platform that unifies patient, physician, hospital, and employer workflows under a single grounded agentic engine. The five-stage

Table 8: Four portals with primary stakeholders and AI agent feature set.

Portal	Primary User	Key Agent Capabilities
Patient	Patient	Plain-language condition summaries (jargon decoder), cost share breakdown, medication guides (DailyMed), journey timeline, financial assistance search, follow-up reminders
Physician	Clinician	Navigator (jargon + insurance + claim routing in parallel), one-tap Tier 2 approval queue, autonomous action feed, prior auth status tracking, formulary checker
Hospital	Admin/Billing	Full RCM state machine (draft→paid/denied→appealed), AR aging dashboard, AI-generated appeal letters, CMS benchmark comparison, HIPAA audit log, revenue forecasting
Employer	HR/Benefits	Claude-powered benefits optimizer (savings with dollar amounts), open enrollment automation, COBRA compliance (44-day window), ACA 1095-C reporting, monthly admin package

pipeline achieves 94.2% NER accuracy, AUC 0.87 on denial prediction, $MRR@5 = 0.91$ on ICD-10 retrieval, and sub-1.8 s end-to-end latency at the 95th percentile. The three-tier autonomy model provides a principled, code-enforced framework for deploying AI agents in high-stakes clinical contexts.

Synthure demonstrates that production-grade multi-agent clinical AI infrastructure can be built on open datasets, public model APIs (HuggingFace, Anthropic), and serverless infrastructure (Vercel, Supabase) for approximately \$5/week in compute costs. The complete system is open-source at github.com/aravinds-kannappan/Synthure.

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Table 9: Database migration overview.

Migration	Contents
001_extensions	uuid-osp, vector (pgvector), pg_cron
002_core	orgs, users, notifications, care_events, audit_logs, realtime_events
003_crm	patients, patient_insurance, providers, payers, communications
004_rcm	claims (state machine), claim_transitions, prior_auths, denial_events, appeals
005_features	encounters, action_queue, discharge_records
006_employer	benefit_plans, enrollments, cobra_events, aca_reports
007_mle	rag_documents (pgvector 384-dim, IVFFlat 200 lists), model_versions, feature_store
008_platform	webhooks, api_keys, billing_usage
009_rls	Row Level Security across all tables

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